Internship Management System - Complete Documentation

> ***Note:*** *This document contains Mermaid diagrams that render in GitHub, GitLab, and most modern markdown viewers. If diagrams don't render in your viewer, consider using a Mermaid-compatible markdown viewer or the Mermaid Live Editor.*

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Introduction

This document provides comprehensive visual documentation of the Internship Management System. It includes entity relationships, user workflows, API endpoints, state transitions, and quick reference guides for developers, stakeholders, and new team members.

The system manages the complete lifecycle of internships from company registration and verification through student applications, internship completion, and credit transfer approval.

Document Purpose:

- Provide visual reference for system architecture and data relationships
- Document all user workflows and their corresponding API endpoints
- Serve as onboarding material for new developers
- Act as a reference for stakeholders understanding system capabilities
- Support maintenance and future development efforts

Glossary

Key Terms:

- **Entity:** A data model representing a core concept in the system (Student, Company, Internship, etc.)
- **Workflow:** A sequence of steps that accomplish a specific business process
- **API Endpoint:** A specific URL path that handles HTTP requests for system operations
- **Status Transition:** A change from one status value to another in an entity's lifecycle
- **Role:** A classification of system users (Student, Company, Mentor, Admin)
- **Cardinality:** The numerical relationship between entities (1:1, 1:N, N:M)
- **Foreign Key (FK):** A field that references another entity's primary key
- **Primary Key (PK):** A unique identifier field for an entity
- **Unique Key (UK):** A field that must be unique across all records

System Overview

The Internship Management System is a three-tier application:

- **Frontend**: Next.js React application with role-based dashboards
- **Backend**: Node.js/Express REST API with business logic
- **Data Layer**: MongoDB database with Redis caching

Key Features

- Multi-role support (Student, Company, Mentor, Admin)
- AI-powered company verification and application ranking
- Multi-stage approval workflows
- Credit transfer management
- Real-time notifications
- Document management and storage
- Analytics and reporting

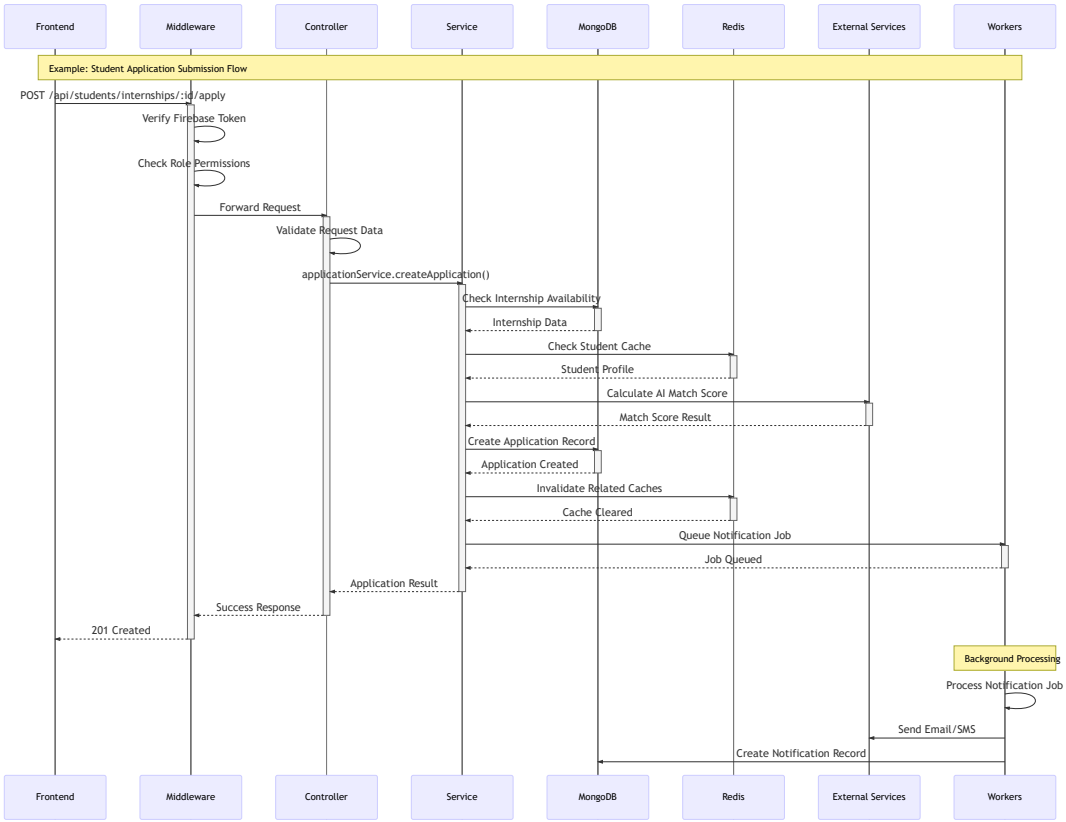
System Architecture

High-Level Architecture Diagram

This diagram shows the complete system architecture with all layers, components, and their interactions. Data flows from the frontend through the backend to the data layer, with external services integrated at the backend level.

Detailed Component Interactions

This sequence diagram illustrates how different system components interact during a typical operation. The example shows a student submitting an application, demonstrating the flow through middleware, controllers, services, database, cache, external services, and background workers.



Architecture Components

Frontend Layer (Next.js Application)

Purpose: Provides user interfaces for all system roles with responsive design and real-time updates.

Key Components:

- **Role-Based Dashboards:** Separate dashboard implementations for Student, Company, Mentor, and Admin roles
- **Shared Components:** Reusable UI components (forms, tables, modals, charts) used across all dashboards
- **State Management:** Zustand for global state management (user session, notifications, cache)
- **API Client:** Axios-based HTTP client with interceptors for authentication and error handling
- **Real-Time Updates:** WebSocket connections for live notifications and status updates
- **Document Management:** File upload components with progress tracking and validation

Technology Stack:

- Next.js 14 (App Router)
- React 18
- TypeScript
- Tailwind CSS
- Zustand (State Management)
- Axios (HTTP Client)

Data Flow:

1. User interacts with dashboard components
2. Components dispatch actions to state management
3. API client makes authenticated HTTP requests to backend
4. Responses update local state and trigger UI re-renders

5. Real-time updates received via WebSocket connections

Backend Layer (Node.js/Express)

Purpose: Handles business logic, data processing, authentication, and orchestrates interactions between frontend, database, and external services.

Key Components:

1. Controllers Layer:

- Route handlers that receive HTTP requests
- Request validation and sanitization
- Response formatting and error handling
- Delegates business logic to service layer
- Examples: `studentController.js`, `companyController.js`, `adminController.js`

2. Services Layer:

- Core business logic implementation
- Data transformation and validation
- Orchestrates database operations
- Integrates with external services
- Examples: `applicationService.js`, `creditService.js`, `internshipService.js`

3. Middleware Layer:

- **Authentication Middleware:** Verifies Firebase tokens and extracts user identity
- **Authorization Middleware:** Checks role-based permissions for routes
- **Validation Middleware:** Validates request payloads against schemas
- **Rate Limiting:** Prevents abuse with request rate limits
- **Error Handling:** Centralized error handling and logging
- **Query Performance:** Monitors and logs slow database queries

4. Background Workers:

- **Email Worker:** Processes email notification jobs
- **SMS Worker:** Sends SMS notifications for critical events
- **Analytics Worker:** Generates periodic analytics snapshots
- **Credit Notification Worker:** Sends reminders for pending credit requests
- **Deadline Reminder Worker:** Notifies users of approaching deadlines
- **AI Tagging Worker:** Processes AI-based tagging for internships
- **Logbook Worker:** Processes and validates logbook entries
- **Report Worker:** Generates PDF reports and certificates

5. Queue Management (BullMQ):

- Job queue for asynchronous task processing
- Retry logic for failed jobs
- Job prioritization and scheduling
- Progress tracking and monitoring

Technology Stack:

- Node.js 18+
- Express.js
- TypeScript/JavaScript
- BullMQ (Job Queue)

- Firebase Admin SDK
- Mongoose (MongoDB ODM)

Data Flow:

1. Request arrives at Express server
2. Middleware chain processes request (auth, validation, rate limiting)
3. Controller receives validated request
4. Controller delegates to appropriate service
5. Service performs business logic and database operations
6. Service may queue background jobs for async processing
7. Service returns result to controller
8. Controller formats response and sends to client
9. Background workers process queued jobs independently

Data Layer

Purpose: Provides persistent storage and caching for application data.

Components:

1. MongoDB Database:

- **Primary Data Store:** All persistent application data
- **Collections:** Students, Companies, Mentors, Admins, Internships, Applications, CreditRequests, InternshipCompletions, Notifications, AiUsageLogs, AnalyticsSnapshots
- **Indexes:** Optimized indexes on frequently queried fields (status, dates, foreign keys)
- **Aggregation Pipelines:** Complex queries for analytics and reporting
- **Transactions:** ACID transactions for critical operations (credit transfers, status updates)

2. Redis Cache & Queue:

- **Caching:** Frequently accessed data (user profiles, internship listings, application counts)
- **Session Storage:** User session data and temporary tokens
- **Job Queue:** BullMQ job queue for background processing
- **Rate Limiting:** Request rate limit counters
- **Real-Time Data:** Temporary storage for real-time features

Technology Stack:

- MongoDB 6.0+
- Redis 7.0+
- Mongoose (ODM)
- Redis Client

Data Flow:

1. Service layer checks Redis cache for requested data
2. If cache hit, return cached data immediately
3. If cache miss, query MongoDB
4. Store result in Redis cache with TTL
5. Return data to service layer
6. Background workers update cache when data changes
7. Job queue stores pending background tasks in Redis

External Services Integration

Purpose: Extends system capabilities with specialized third-party services.

Services:

1. Firebase Authentication:

- **User Management:** Create, update, delete user accounts
- **Token Verification:** Verify JWT tokens on every API request
- **Password Management:** Handle password resets and email verification
- **Multi-Factor Authentication:** Optional 2FA for admin accounts
- **Integration Point:** Middleware layer verifies tokens before processing requests

2. Cloudflare R2 Storage:

- **Document Storage:** Company verification documents, student resumes, certificates
- **Public Access:** Configured for public read access on approved documents
- **Presigned URLs:** Temporary URLs for secure document access
- **Integration Point:** Service layer uploads/downloads documents via R2 SDK

3. Email Service (SMTP):

- **Transactional Emails:** Application updates, credit approvals, verification results
- **Notification Emails:** Daily digests, reminders, system announcements
- **Templates:** HTML email templates for consistent branding
- **Integration Point:** Email worker processes email jobs from queue

4. Gemini AI API:

- **Company Verification:** Analyzes company documents for legitimacy
- **Application Ranking:** Calculates match scores between students and internships
- **Skill Gap Analysis:** Identifies skill gaps and recommends learning modules
- **Interview Practice:** Powers AI interview bot for student practice
- **Integration Point:** Service layer makes API calls for AI operations

5. SMS Service:

- **Critical Notifications:** High-priority alerts sent via SMS
- **Two-Factor Authentication:** SMS-based 2FA codes
- **Integration Point:** SMS worker processes SMS jobs from queue

Technology Stack:

- Firebase Admin SDK
- AWS SDK (for R2 compatible API)
- Nodemailer (SMTP)
- Google Generative AI SDK
- Twilio/SMS Gateway API

Data Flow:

1. Service layer identifies need for external service
2. For synchronous operations (AI, auth), service calls external API directly
3. For asynchronous operations (email, SMS), service queues background job
4. Background worker processes job and calls external service
5. Results stored in database and cache as needed
6. User notified of completion via notification system

Architecture Patterns

1. Three-Tier Architecture:

- Clear separation between presentation (frontend), business logic (backend), and data (database)
- Each tier can be scaled independently
- Changes in one tier have minimal impact on others

2. Service-Oriented Architecture:

- Business logic organized into focused services
- Each service handles a specific domain (applications, credits, internships)
- Services can be extracted into microservices if needed

3. Queue-Based Processing:

- Long-running tasks processed asynchronously
- Improves API response times
- Provides retry logic and fault tolerance

4. Caching Strategy:

- Cache-aside pattern for frequently accessed data
- Cache invalidation on data updates
- TTL-based expiration for stale data prevention

5. Role-Based Access Control (RBAC):

- Permissions checked at middleware level
- Fine-grained access control per endpoint
- Role hierarchy (Student < Company < Mentor < Admin < Super Admin)

Scalability Considerations

Horizontal Scaling:

- Frontend: Deploy multiple Next.js instances behind load balancer
- Backend: Deploy multiple Express instances with shared Redis/MongoDB
- Workers: Scale worker processes independently based on queue depth

Vertical Scaling:

- Database: Increase MongoDB instance size for larger datasets
- Cache: Increase Redis memory for larger cache

Performance Optimizations:

- Database indexes on frequently queried fields
- Redis caching for hot data
- Pagination for large result sets
- Lazy loading for frontend components
- CDN for static assets

Monitoring & Observability:

- Application logs stored and analyzed
- Performance metrics tracked (response times, error rates)
- Queue depth monitoring for worker scaling
- Database query performance monitoring

Entity Relationship Diagram

Complete ERD

This diagram shows all core entities in the Internship Management System, their relationships, and key attributes. The system uses MongoDB, so primary keys are ObjectId types, and relationships are implemented through foreign key references.

Relationship Descriptions

This section describes the cardinality and nature of relationships between entities in the system.

Student Relationships:

- One student can have many applications (1:N) - Students apply to multiple internships
- One student can have many credit requests (1:N) - Students submit multiple credit requests over time
- One student can have many internship completions (1:N) - Students complete multiple internships
- Many students are assigned to one mentor (N:1) - Mentors guide multiple students
- One student receives many notifications (1:N) - System sends notifications for various events

Company Relationships:

- One company can post many internships (1:N) - Companies create multiple internship postings
- One company receives many applications (1:N) - Companies receive applications across all their internships
- One company receives many notifications (1:N) - System sends notifications for application updates, verifications, etc.

Internship Relationships:

- One internship has many applications (1:N) - Multiple students apply to each internship
- One internship results in many completions (1:N) - Multiple students can complete the same internship (different time periods)

Application Relationships:

- One application leads to one completion (1:1) - Each accepted application results in one completion record

InternshipCompletion Relationships:

- One completion generates one credit request (1:1) - Each completion creates exactly one credit request

Mentor Relationships:

- One mentor reviews many applications (1:N) - Mentors review applications from their assigned students
- One mentor reviews many credit requests (1:N) - Mentors review credit requests from their students
- One mentor receives many notifications (1:N) - System sends notifications for pending reviews

Admin Relationships:

- Admins verify companies (1:N) - Admins review and verify multiple company registrations
- Admins approve internships (1:N) - Admins review and approve multiple internship postings
- Admins approve credit requests (1:N) - Admins perform final approval on multiple credit requests
- Admins receive many notifications (1:N) - System sends notifications for pending verifications and approvals

Notification Relationships:

- One user (any role) receives many notifications (1:N) - Notifications are sent to users across all roles
- Notifications are polymorphic - The userId field can reference Student, Company, Mentor, or Admin based on the role field

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Data Models

Student

Purpose: Primary entity for learners in the system

Key Fields:

- `studentId` : Unique identifier (e.g., "STU-2024-001")
- `firebaseUid` : Firebase authentication ID
- `email` : Student email address
- `profile` : Personal information, skills, education
- `readinessScore` : AI-calculated readiness (0-100)
- `completedModules` : Array of completed training modules
- `credits` : Object tracking earned and pending credits
- `appliedInternships` : Array of internship IDs applied to
- `completedInternships` : Count of completed internships
- `status` : active, inactive, suspended

Company

Purpose: Organizations posting internships

Key Fields:

- `companyId` : Unique identifier (e.g., "COM-2024-001")
- `firebaseUid` : Firebase authentication ID
- `companyName` : Official company name
- `email` : Company contact email
- `documents` : Verification documents (registration, tax, etc.)
- `pointOfContact` : Contact person details
- `status` : pending_verification, verified, rejected, blocked, reappeal
- `aiVerification` : AI verification results
- `adminReview` : Admin review details and history
- `reappeal` : Reappeal submission data
- `stats` : Performance metrics (internships posted, completion rate)

Mentor

Purpose: Faculty members who guide students

Key Fields:

- `mentorId` : Unique identifier (e.g., "MEN-2024-001")
- `firebaseUid` : Firebase authentication ID
- `email` : Mentor email address
- `profile` : Name, department, expertise
- `assignedStudents` : Array of student IDs
- `workload` : Current and maximum student capacity
- `status` : active, inactive, on_leave

Admin

Purpose: System administrators with various permission levels

Key Fields:

- `adminId` : Unique identifier (e.g., "ADM-2024-001")
- `firebaseUid` : Firebase authentication ID
- `email` : Admin email address
- `name` : Admin full name
- `role` : super_admin, admin, support
- `permissions` : Array of permission strings
- `status` : active, inactive

Internship

Purpose: Job postings created by companies

Key Fields:

- `internshipId` : Unique identifier (e.g., "INT-2024-001")
- `companyId` : Reference to company
- `title` : Internship title
- `description` : Detailed description
- `department` : Target department (CS, IT, Business, etc.)
- `requiredSkills` : Array of required skills
- `startDate` : Internship start date
- `applicationDeadline` : Application deadline
- `slots` : Total available positions
- `slotsRemaining` : Remaining available positions
- `status` : draft, pending_admin_verification, admin_approved, pending_mentor_verification, open_for_applications, closed, cancelled
- `adminReview` : Admin review details
- `mentorApproval` : Mentor approval details
- `departmentApprovals` : Array of department approvals
- `aiTags` : AI-generated tags and categorization
- `isDeleted` : Soft delete flag

Application

Purpose: Student applications to internships

Key Fields:

- `applicationId` : Unique identifier (e.g., "APP-2024-001")

- `studentId` : Reference to student
- `internshipId` : Reference to internship
- `companyId` : Reference to company
- `department` : Application department
- `status` : pending, mentor_approved, mentor_rejected, shortlisted, rejected, accepted, withdrawn, completed
- `appliedAt` : Application submission timestamp
- `coverLetter` : Student cover letter
- `mentorApproval` : Mentor review details
- `companyFeedback` : Company feedback
- `aiRanking` : AI-calculated match score
- `timeline` : Array of status change events

InternshipCompletion

Purpose: Records completed internships and serves as the basis for credit transfer requests

Key Fields:

- `completionId` : Unique identifier (e.g., "CMP-2024-001")
- `studentId` : Reference to student who completed the internship
- `internshipId` : Reference to the internship that was completed
- `companyId` : Reference to the company where internship was completed
- `totalHours` : Total hours worked during the internship
- `creditsEarned` : Credits to be awarded (calculated based on hours and quality)
- `completionDate` : Date when internship was marked complete
- `evaluation` : Company evaluation object containing ratings and feedback
- `certificates` : Completion certificates and supporting documents
- `status` : pending, completed, verified
- `creditRequest` : Reference to associated credit request
- `companyCompletion` : Company confirmation details (timestamp, confirmed by)

Completion Process:

1. Company marks internship as complete
2. System creates InternshipCompletion record
3. Company provides evaluation and feedback
4. System calculates eligible credits
5. Student can then submit credit request

CreditRequest

Purpose: Multi-stage approval workflow for credit transfer

Key Fields:

- `creditRequestId` : Unique identifier (e.g., "CR-2024-001")
- `studentId` : Reference to student
- `internshipCompletionId` : Reference to completion
- `internshipId` : Reference to internship
- `mentorId` : Assigned mentor for review
- `requestedCredits` : Credits requested by student
- `calculatedCredits` : System-calculated credits based on hours and quality

- `status` : pending_mentor_review, mentor_approved, mentor_rejected, pending_admin_review, admin_approved, admin_rejected, credits_added, completed
- `mentorReview` : Mentor review details (comments, decision, timestamp)
- `adminReview` : Admin review details (comments, decision, timestamp)
- `submissionHistory` : Array of submission attempts with timestamps
- `certificate` : Generated certificate document reference
- `requestedAt` : Request submission timestamp

Credit Calculation: Credits are calculated based on:

- Total internship hours completed
- Quality of work (from company evaluation)
- Logbook completeness and quality
- Compliance with NEP 2020 guidelines

Notification

Purpose: System notifications sent to users across all roles

Key Fields:

- `notificationId` : Unique identifier (e.g., "NOT-2024-001")
- `userId` : Reference to user (polymorphic - can be Student, Company, Mentor, or Admin)
- `role` : User role (student, mentor, admin, company)
- `type` : Notification type (application_update, credit_approved, verification_complete, etc.)
- `title` : Notification title
- `message` : Notification message content
- `priority` : low, medium, high, critical
- `actionUrl` : Optional URL for user action
- `read` : Boolean flag indicating if notification has been read
- `readAt` : Timestamp when notification was read
- `deliveries` : Array of delivery attempts across channels (email, sms, whatsapp, push)
- `metadata` : Additional context-specific data

Notification Types:

- Application updates (submitted, approved, rejected, shortlisted)
- Credit request updates (submitted, approved, needs revision)
- Company verification updates (verified, rejected, reappeal status)
- Internship updates (approved, rejected, deadline approaching)
- System announcements and reminders

Multi-Channel Delivery: Notifications can be delivered through multiple channels:

- In-app notifications (always)
- Email (configurable per notification type)
- SMS (for critical notifications)
- WhatsApp (optional)
- Push notifications (mobile apps)

User Workflows

Student Journey

Student Journey Steps:

1. **Registration & Profile Setup**
   - Student creates account with email
   - Completes profile with skills, education, interests
   - System calculates readiness score
2. **Browse & Apply**
   - Browse available internships
   - Filter by department, skills, location
   - View AI match scores
   - Submit application with cover letter
3. **Mentor Review**
   - Mentor reviews application
   - Checks student readiness and fit
   - Approves or rejects with feedback
4. **Company Review**
   - Company reviews approved applications
   - Shortlists candidates
   - Provides feedback on rejections
5. **Acceptance**
   - Student accepts offer
   - Application status updated to accepted
6. **Internship Completion**
   - Student completes internship
   - Company marks completion
   - System creates completion record
7. **Credit Request**
   - Student submits credit request
   - Mentor reviews logbook and quality
   - Admin performs final approval
   - Credits added to student account

Company Journey

Company Journey Steps:

1. **Registration & Verification**
   - Company creates account
   - Uploads verification documents (registration, tax ID, etc.)
   - AI performs initial verification
   - Admin reviews and approves/rejects
   - Option to submit reappeal if rejected
2. **Post Internship**
   - Create internship posting with details
   - Specify requirements, skills, duration
   - Submit for admin review
3. **Admin Approval**
   - Admin checks compliance
   - Verifies company is verified
   - Approves or requests revisions
4. **Mentor Approval**
   - Mentor verifies department fit
   - Checks requirements alignment
   - Approves for student applications
5. **Application Management**
   - Review incoming applications
   - View AI match scores
   - Shortlist or reject candidates
   - Provide feedback
6. **Intern Management**
   - Monitor intern progress
   - Review logbook entries
   - Mark completion when finished
   - Provide evaluation and feedback

Mentor Journey

Mentor Journey Steps:

1. **Dashboard Overview**
   - View pending tasks count
   - See assigned students
   - Check workload metrics
2. **Application Review**
   - Review applications from assigned students
   - Check student profile and readiness score
   - View AI match score
   - Approve or reject with feedback
   - Forward approved applications to company
3. **Credit Request Review**
   - Review credit requests from students
   - Check internship completion details
   - Review logbook entries
   - Verify quality criteria met
   - Approve for admin review or request revisions
4. **Internship Posting Review**
   - Review internship postings for department
   - Verify requirements and fit
   - Approve or reject with feedback
   - Approved postings become available to students

Admin Journey

Admin Journey Steps:

1. **Dashboard Overview**
   - View system-wide metrics

- See pending verification counts
- Monitor system health

2. **Company Verification**
   - Review company registration requests
   - Check AI verification results
   - Review submitted documents
   - Verify company legitimacy
   - Approve or reject with detailed feedback
   - Handle reappeal submissions
3. **Internship Approval**
   - Review internship postings
   - Verify compliance with policies
   - Check company verification status
   - Approve for mentor review or reject
   - Monitor internship lifecycle
4. **Credit Request Final Approval**
   - Review mentor-approved credit requests
   - Verify compliance and quality
   - Check calculation accuracy
   - Perform final approval
   - System adds credits to student account
   - Generate completion certificate
5. **Analytics & Reporting**
   - View system analytics
   - Monitor trends and metrics
   - Export reports for stakeholders
   - Track performance indicators

State Transition Diagrams

Internship Status Flow

Internship Status Transitions:

This table describes all valid status transitions for internships, including who can trigger each transition and under what conditions.

From Status	To Status	Trigger	Role	Conditions

draft	pending_admin_verification	Company submits posting	Company	All required fields completed
pending_admin_verification	admin_approved	Admin approves	Admin	Company is verified, posting meets compliance
pending_admin_verification	admin_rejected	Admin rejects	Admin	Compliance issues or company not verified
admin_approved	pending_mentor_verification	Automatic	System	Immediately after admin approval
pending_mentor_verification	open_for_applications	Mentor approves	Mentor	Department fit verified
pending_mentor_verification	mentor_rejected	Mentor rejects	Mentor	Requirements don't match department
open_for_applications	closed	Deadline passed or slots filled	System	Application deadline reached or all slots filled
open_for_applications	cancelled	Company cancels	Company	Company decides to cancel posting

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Application Status Flow

Application Status Transitions:

This table describes all valid status transitions for applications throughout the approval workflow.

From Status	To Status	Trigger	Role	Conditions
-	pending	Student submits application	Student	Internship is open, student hasn't already applied

pending	mentor_approved	Mentor approves	Mentor	Student meets requirements, good fit
pending	mentor_rejected	Mentor rejects	Mentor	Student doesn't meet requirements or poor fit
mentor_approved	shortlisted	Company shortlists	Company	Candidate meets company criteria
mentor_approved	rejected	Company rejects	Company	Candidate doesn't meet company needs
shortlisted	accepted	Student accepts offer	Student	Student chooses to accept the offer
shortlisted	withdrawn	Student withdraws	Student	Student no longer interested
accepted	completed	Internship completed	Company	Internship period finished successfully

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Credit Request Status Flow

Credit Request Status Transitions:

This table describes the multi-stage approval workflow for credit transfer requests.

From Status	To Status	Trigger	Role	Conditions
-	pending_mentor_review	Completion marked	System	InternshipCompletion record created
pending_mentor_review	mentor_approved	Mentor approves	Mentor	Logbook quality meets standards, hours verified
pending_mentor_review	mentor_rejected	Mentor rejects	Mentor	Quality issues or insufficient documentation
mentor_rejected	pending_student_action	Needs revision	System	Automatic after mentor rejection
pending_student_action	pending_mentor_review	Student resubmits	Student	Student addresses feedback and resubmits
mentor_approved	pending_admin_review	Automatic	System	Immediately after mentor approval

pending_admin_review	admin_approved	Admin approves	Admin	Compliance verified, calculations correct
pending_admin_review	admin_rejected	Admin rejects	Admin	Compliance issues or calculation errors
admin_rejected	pending_student_action	Needs revision	System	Automatic after admin rejection
admin_approved	credits_added	System processes	System	Credits added to student account
credits_added	completed	Certificate generated	System	Certificate PDF generated and stored

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Company Verification Flow

Company Verification Status Transitions:

This table describes the company verification and reappeal workflow.

From Status	To Status	Trigger	Role	Conditions
-	pending_verification	Company registers	System	Account created, documents uploaded
pending_verification	verified	Admin approves	Admin	AI verification passed, documents legitimate
pending_verification	rejected	Admin rejects	Admin	AI verification failed or documents invalid
rejected	reappeal	Company submits reappeal	Company	Additional documents provided within timeframe
reappeal	verified	Reappeal approved	Admin	New evidence satisfactory
reappeal	blocked	Reappeal rejected	Admin	Evidence still insufficient, permanently blocked
verified	suspended	Policy violation	Admin	Minor policy violation, temporary suspension
suspended	verified	Issue resolved	Admin	Company addresses violation

verified	blocked	Serious violation	Admin	Major policy violation, permanent block

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API Endpoint Reference

Authentication Endpoints

These endpoints are used across all user roles for authentication and profile management.

HTTP Method	Endpoint	Description	Auth Required	Permissions
POST	`/api/auth/students/register`	Register new student account	No	Public
POST	`/api/auth/companies/register`	Register new company account	No	Public
POST	`/api/auth/mentors/register`	Register new mentor account	No	Public
POST	`/api/auth/admins/register`	Register new admin account	No	Super Admin (production)
POST	`/api/auth/login`	User login with Firebase token	No	Public
POST	`/api/auth/send-password-reset`	Send password reset email	No	Public
POST	`/api/auth/verify-email`	Send email verification link	Yes	All authenticated users
GET	`/api/auth/verify-email`	Verify email with code	No	Public
GET	`/api/auth/me`	Get current user profile	Yes	All authenticated users
PATCH	`/api/auth/me`	Update user profile	Yes	All authenticated users
POST	`/api/auth/password`	Change password	Yes	All authenticated users
POST	`/api/auth/profile/image`	Upload profile image	Yes	All authenticated users
POST	`/api/auth/profile/resume`	Upload resume document	Yes	Student
DELETE	`/api/auth/account`	Delete account (soft delete)	Yes	All authenticated users

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Notification Endpoints

These endpoints are used across all user roles for managing notifications.

HTTP Method	Endpoint	Description	Auth Required	Permissions
GET	`/api/notifications`	Get user notifications with pagination	Yes	All authenticated users
PATCH	`/api/notifications/:id/read`	Mark specific notification as read	Yes	All authenticated users
PATCH	`/api/notifications/read-all`	Mark all notifications as read	Yes	All authenticated users

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Student Endpoints

HTTP Method	Endpoint	Description	Auth Required	Pe
GET	`/api/students/dashboard`	Get student dashboard with mentor info	Yes	Stu
GET	`/api/students/profile`	Get student profile with credits	Yes	Stu
GET	`/api/students/internships`	Browse available internships with filters	Yes	Stu
GET	`/api/students/internships/recommended`	Get AI-recommended internships	Yes	Stu
GET	`/api/students/internships/completed`	Get completed internships with credit status	Yes	Stu
GET	`/api/students/internships/:internshipId`	Get internship details with match analysis	Yes	Stu
POST	`/api/students/internships/:internshipId/apply`	Apply to internship with cover letter	Yes	Stu
GET	`/api/students/applications`	Get student's applications	Yes	Stu

		with filters		
GET	`/api/students/applications/:applicationId`	Get application details and timeline	Yes	Stu
DELETE	`/api/students/applications/:applicationId`	Withdraw application	Yes	Stu
GET	`/api/students/modules/recommended`	Get recommended learning modules	Yes	Stu
POST	`/api/students/modules/start`	Start a learning module	Yes	Stu
POST	`/api/students/modules/complete`	Complete a learning module	Yes	Stu
POST	`/api/students/interviews/start`	Start AI interview practice session	Yes	Stu
POST	`/api/students/interviews/answer`	Submit interview answer for AI feedback	Yes	Stu
POST	`/api/students/interviews/end`	End interview practice session	Yes	Stu
GET	`/api/students/interviews/history`	Get interview practice history	Yes	Stu
POST	`/api/students/logbooks`	Submit logbook entry	Yes	Stu
GET	`/api/students/logbooks`	Get student's logbook entries	Yes	Stu
GET	`/api/students/credits`	Get credits summary	Yes	Stu
POST	`/api/students/reports/nep`	Generate NEP compliance report	Yes	Stu
POST	`/api/students/chatbot`	Query AI chatbot for	Yes	Stu

		guidance		
POST	`/api/students/:studentId/credit-requests`	Create new credit request	Yes	Stu
GET	`/api/students/:studentId/credit-requests`	Get all credit requests for student	Yes	Stu
GET	`/api/students/:studentId/credit-requests/:requestId`	Get credit request details	Yes	Stu
PUT	`/api/students/:studentId/credit-requests/:requestId/resubmit`	Resubmit rejected credit request	Yes	Stu
GET	`/api/students/:studentId/credit-requests/:requestId/status`	Get real-time credit request status	Yes	Stu
POST	`/api/students/:studentId/credit-requests/:requestId/reminder`	Send reminder to current reviewer	Yes	Stu
GET	`/api/students/:studentId/credits/history`	Get credit history	Yes	Stu
GET	`/api/students/:studentId/credits/certificate/:requestId`	Download credit transfer certificate	Yes	Stu

Student Workflow Mapping:

This section maps each step in the student journey (see Student Journey diagram) to the specific API endpoints used.

1. Registration & Profile Setup

Workflow Steps: Student Registers → Complete Profile

Step	Endpoint	Method	Description
Register account	`/api/auth/students/register`	POST	Create new student account with email and password
Login	`/api/auth/login`	POST	Authenticate with Firebase token
Verify email	`/api/auth/verify-email`	POST	Send email verification link
Confirm email	`/api/auth/verify-email`	GET	Verify email with code from link
View profile	`/api/students/profile`	GET	Get student profile with readiness score
Update profile	`/api/auth/me`	PATCH	Update personal information, skills, education
Upload profile image	`/api/auth/profile/image`	POST	Upload profile picture

Upload resume	`/api/auth/profile/resume`	POST	Upload resume document
View dashboard	`/api/students/dashboard`	GET	View dashboard with mentor info and metrics

2. Browse & Apply to Internships

Workflow Steps: Browse Internships → Find Suitable Internship → Submit Application

Step	Endpoint	Method	Description
Browse internships	`/api/students/internships`	GET	List available internships with filters (department, skills, location)
Get recommendations	`/api/students/internships/recommended`	GET	Get AI-recommended internships based on profile
View internship details	`/api/students/internships/:internshipId`	GET	View full internship details with AI match score
Submit application	`/api/students/internships/:internshipId/apply`	POST	Apply to internship with cover letter

3. Track Applications

Workflow Steps: Wait for Mentor Review → Wait for Company Review → Accept Offer

Step	Endpoint	Method	Description
View all applications	`/api/students/applications`	GET	List all applications with status and filters
Check application status	`/api/students/applications/:applicationId`	GET	View application details, timeline, and feedback
Withdraw application	`/api/students/applications/:applicationId`	DELETE	Withdraw pending or shortlisted application
Receive notifications	`/api/notifications`	GET	Get notifications about application status changes

4. Learning & Development

Workflow Steps: Complete training modules, practice interviews, get AI guidance

Step	Endpoint	Method	Description
Get recommended modules	`/api/students/modules/recommended`	GET	Get personalized learning module recommendations
Start module	`/api/students/modules/start`	POST	Begin a learning module

Complete module	`/api/students/modules/complete`	POST	Mark module as complete and update readiness score
Start interview practice	`/api/students/interviews/start`	POST	Begin AI-powered interview practice session
Submit interview answer	`/api/students/interviews/answer`	POST	Submit answer and receive AI feedback
End interview session	`/api/students/interviews/end`	POST	Complete interview practice session
View interview history	`/api/students/interviews/history`	GET	View past interview practice sessions
Get AI guidance	`/api/students/chatbot`	POST	Ask AI chatbot for career guidance

5. Internship Completion & Credits

Workflow Steps: Complete Internship → Company Marks Complete → Submit Credit Request → Mentor Reviews → Admin Reviews → Credits Added

Step	Endpoint	Method	Description
View completed internships	`/api/students/internships/completed`	GET	List completed internships with credit request status
Submit logbook entry	`/api/students/logbooks`	POST	Submit daily/weekly logbook entry during internship
View logbook entries	`/api/students/logbooks`	GET	View all logbook entries
Create credit request	`/api/students/:studentId/credit-requests`	POST	Submit credit transfer request after completion
View all credit requests	`/api/students/:studentId/credit-requests`	GET	List all credit requests with status
View credit request details	`/api/students/:studentId/credit-requests/:requestId`	GET	View detailed

			credit request with reviews
Track credit status	`/api/students/:studentId/credit-requests/:requestId/status`	GET	Get real-time status of credit request
Resubmit if rejected	`/api/students/:studentId/credit-requests/:requestId/resubmit`	PUT	Resubmit credit request with revisions
Send reminder	`/api/students/:studentId/credit-requests/:requestId/reminder`	POST	Send reminder to current reviewer
View credit history	`/api/students/:studentId/credits/history`	GET	View all earned credits history
Download certificate	`/api/students/:studentId/credits/certificate/:requestId`	GET	Download credit transfer certificate PDF
View credits summary	`/api/students/credits`	GET	View total earned and pending credits
Generate NEP report	`/api/students/reports/nep`	POST	Generate NEP compliance report

6. Notifications

Workflow Steps: Receive and manage notifications throughout all workflows

Step	Endpoint	Method	Description
View notifications	`/api/notifications`	GET	Get paginated list of notifications
Mark notification as read	`/api/notifications/:id/read`	PATCH	Mark specific notification as read
Mark all as read	`/api/notifications/read-all`	PATCH	Mark all notifications as read

Company Endpoints

HTTP	Endpoint	Description	Auth	Perm

Method			Required	
GET	`/api/companies/dashboard`	Get company dashboard with metrics	Yes	Compar
GET	`/api/companies/profile`	Get company profile	Yes	Compar
PATCH	`/api/companies/profile`	Update company profile	Yes	Compar
POST	`/api/companies/documents`	Upload verification documents	Yes	Compar
GET	`/api/companies/status`	Get verification status	Yes	Compar
POST	`/api/companies/re-appeal`	Re-appeal rejection (legacy)	Yes	Compar (rejecte
POST	`/api/companies/reappeal`	Submit reappeal request with documents	Yes	Compar (rejecte
GET	`/api/companies/reappeal/status`	Get reappeal status	Yes	Compar
POST	`/api/companies/internships`	Create new internship posting	Yes	Compar (verified
GET	`/api/companies/internships`	Get all company internships	Yes	Compar
GET	`/api/companies/internships/:internshipId`	Get internship details	Yes	Compar
PATCH	`/api/companies/internships/:internshipId`	Update internship posting	Yes	Compar
DELETE	`/api/companies/internships/:internshipId`	Cancel/delete internship	Yes	Compar
POST	`/api/companies/internships/:internshipId/complete`	Mark internship as complete	Yes	Compar
GET	`/api/companies/internships/:internshipId/applicants`	Get applicants	Yes	Compar

		with filters		
GET	`/api/companies/internships/:internshipId/metrics`	Get internship-specific metrics	Yes	Compar
GET	`/api/companies/applications`	Get all applications to company internships	Yes	Compar
GET	`/api/companies/applications/:applicationId`	Get application details	Yes	Compar
POST	`/api/companies/applications/review`	Review applications (bulk)	Yes	Compar
POST	`/api/companies/applications/shortlist`	Shortlist candidates (bulk)	Yes	Compar
POST	`/api/companies/applications/reject`	Reject candidates (bulk)	Yes	Compar
POST	`/api/companies/applications/approve`	Approve single application	Yes	Compar
POST	`/api/companies/applications/reject-single`	Reject single application	Yes	Compar
GET	`/api/companies/interns`	Get active interns	Yes	Compar
GET	`/api/companies/interns/progress`	Get intern progress tracking	Yes	Compar
GET	`/api/companies/interns/:studentId/logbooks`	Get intern's logbook entries	Yes	Compar
POST	`/api/companies/interns/:applicationId/complete`	Mark internship as complete	Yes	Compar
POST	`/api/companies/logbooks/:logbookId/feedback`	Provide logbook feedback	Yes	Compar
PUT	`/api/companies/completions/:completionId/mark-complete`	Mark completion	Yes	Compar

		with evaluation		
GET	`/api/companies/completions/completed`	Get list of completed internships	Yes	Compar
POST	`/api/companies/events`	Create company event	Yes	Compar
POST	`/api/companies/challenges`	Create coding challenge	Yes	Compar
GET	`/api/companies/analytics`	Get company analytics with date range	Yes	Compar
GET	`/api/companies/analytics/export`	Export analytics report (CSV/PDF)	Yes	Compar

Company Workflow Mapping:

This section maps each step in the company journey (see Company Journey diagram) to the specific API endpoints used.

1. Registration & Verification

Workflow Steps: Company Registers → Submit Documents → AI Verification → Admin Review → Account Verified/Rejected

Step	Endpoint	Method	Description
Register account	`/api/auth/companies/register`	POST	Create new company account with business details
Login	`/api/auth/login`	POST	Authenticate with Firebase token
Upload verification documents	`/api/companies/documents`	POST	Upload registration, tax ID, and other verification documents
Check verification status	`/api/companies/status`	GET	Check current verification status and AI results
Submit reappeal	`/api/companies/reappeal`	POST	Submit reappeal with additional documents if rejected
Check reappeal status	`/api/companies/reappeal/status`	GET	Check status of reappeal submission
Receive notifications	`/api/notifications`	GET	Get notifications about verification status changes

2. Profile Management

Workflow Steps: Update company information and branding

Step	Endpoint	Method	Description
View profile	`/api/companies/profile`	GET	View company profile and verification details
Update profile	`/api/companies/profile`	PATCH	Update company information and point of contact
Update account info	`/api/auth/me`	PATCH	Update email and account settings
Upload profile image	`/api/auth/profile/image`	POST	Upload company logo

3. Internship Management

Workflow Steps: Post Internship → Admin Reviews → Mentor Reviews → Internship Live

Step	Endpoint	Method	Description
View dashboard	`/api/companies/dashboard`	GET	View dashboard with metrics and pending tasks
Create internship posting	`/api/companies/internships`	POST	Create new internship with requirements and details
View all internships	`/api/companies/internships`	GET	List all company internships with status filters
View internship details	`/api/companies/internships/:internshipId`	GET	View full internship details and approval status
Update internship	`/api/companies/internships/:internshipId`	PATCH	Update internship details (if not yet approved)
Delete/cancel internship	`/api/companies/internships/:internshipId`	DELETE	Cancel internship posting
Mark internship complete	`/api/companies/internships/:internshipId/complete`	POST	Mark internship period as complete
View internship metrics	`/api/companies/internships/:internshipId/metrics`	GET	View application and acceptance metrics

4. Application Management

Workflow Steps: Review Applications → Shortlist Candidates → Student Accepts

Step	Endpoint	Method	Description
View all applications	`/api/companies/applications`	GET	List all applications across all internships
View applicants for internship	`/api/companies/internships/:internshipId/applicants`	GET	View applicants for specific internship with filters
View application details	`/api/companies/applications/:applicationId`	GET	View full application with student profile and AI ranking
Approve single application	`/api/companies/applications/approve`	POST	Shortlist a single candidate
Reject single application	`/api/companies/applications/reject-single`	POST	Reject a single application with feedback
Bulk shortlist candidates	`/api/companies/applications/shortlist`	POST	Shortlist multiple candidates at once
Bulk reject candidates	`/api/companies/applications/reject`	POST	Reject multiple applications with feedback
Review applications (bulk)	`/api/companies/applications/review`	POST	Bulk review and update application statuses

5. Intern Management

Workflow Steps: Monitor Intern Progress → Mark Completion → Provide Evaluation

Step	Endpoint	Method	Description
View active interns	`/api/companies/interns`	GET	List all currently active interns
Track intern progress	`/api/companies/interns/progress`	GET	View progress tracking for all

			interns
View intern logbooks	`/api/companies/interns/:studentId/logbooks`	GET	View logbook entries for specific intern
Provide logbook feedback	`/api/companies/logbooks/:logbookId/feedback`	POST	Provide feedback on logbook entry
Mark internship complete	`/api/companies/interns/:applicationId/complete`	POST	Mark internship as complete for student
Submit completion evaluation	`/api/companies/completions/:completionId/mark-complete`	PUT	Submit final evaluation and completion details
View completed internships	`/api/companies/completions/completed`	GET	List all completed internships

6. Analytics & Reporting

Workflow Steps: View performance metrics and export reports

Step	Endpoint	Method	Description
View analytics dashboard	`/api/companies/analytics`	GET	View company analytics with date range filters
Export analytics report	`/api/companies/analytics/export`	GET	Export analytics as CSV or PDF

7. Events & Challenges (Optional Features)

Workflow Steps: Create company events and coding challenges

Step	Endpoint	Method	Description
Create company event	`/api/companies/events`	POST	Create networking or recruitment event
Create coding challenge	`/api/companies/challenges`	POST	Create coding challenge for students

8. Notifications

Workflow Steps: Receive and manage notifications throughout all workflows

Step	Endpoint	Method	Description
View notifications	`/api/notifications`	GET	Get paginated list of notifications
Mark notification as read	`/api/notifications/:id/read`	PATCH	Mark specific notification as read
Mark all as read	`/api/notifications/read-all`	PATCH	Mark all notifications as read

Mentor Endpoints

HTTP Method	Endpoint	Description	Auth Required	Permissi
GET	`/api/mentors/dashboard`	Get mentor dashboard with metrics	Yes	Mentor
GET	`/api/mentor/profile`	Get mentor profile	Yes	Mentor
PUT	`/api/mentor/profile`	Update mentor profile	Yes	Mentor
GET	`/api/mentors/students`	Get assigned students (legacy)	Yes	Mentor
GET	`/api/mentor/students/list`	Get assigned students with filters	Yes	Mentor
GET	`/api/mentor/students/:studentId/details`	Get student details with history	Yes	Mentor
GET	`/api/mentor/students/:studentId/applications`	Get student's applications	Yes	Mentor
GET	`/api/mentors/students/:studentId/progress`	Get student progress tracking	Yes	Mentor
GET	`/api/mentors/applications/pending`	Get pending applications queue	Yes	Mentor
GET	`/api/mentors/applications/:applicationId`	Get application details	Yes	Mentor
POST	`/api/mentors/applications/:applicationId/approve`	Approve application	Yes	Mentor
POST	`/api/mentors/applications/:applicationId/reject`	Reject application with feedback	Yes	Mentor
GET	`/api/mentor/internships/pending`	Get pending internships for approval	Yes	Mentor
GET	`/api/mentor/internships`	Get all internships with filtering	Yes	Mentor
GET	`/api/mentor/internships/:internshipId`	Get internship	Yes	Mentor

		details for review		
POST	`/api/mentor/internships/:internshipId/approve`	Approve internship for department	Yes	Mentor
POST	`/api/mentor/internships/:internshipId/reject`	Reject internship with reasons	Yes	Mentor
GET	`/api/mentors/logbooks/pending`	Get pending logbooks for review	Yes	Mentor
GET	`/api/mentors/logbooks/:logbookId`	Get logbook details	Yes	Mentor
POST	`/api/mentors/logbooks/:logbookId/approve`	Approve logbook entry	Yes	Mentor
POST	`/api/mentors/logbooks/:logbookId/revision`	Request logbook revision	Yes	Mentor
GET	`/api/mentors/credits/pending`	Get pending credit requests (legacy)	Yes	Mentor
POST	`/api/mentors/credits/:requestId/decide`	Approve/reject credit request (legacy)	Yes	Mentor
GET	`/api/mentors/:mentorId/credit-requests/pending`	Get pending credit requests	Yes	Mentor
GET	`/api/mentors/:mentorId/credit-requests/:requestId`	Get credit request details	Yes	Mentor
POST	`/api/mentors/:mentorId/credit-requests/:requestId/review`	Submit mentor review (approve/reject)	Yes	Mentor
GET	`/api/mentors/:mentorId/credit-requests/history`	Get mentor's review history	Yes	Mentor
GET	`/api/mentors/:mentorId/credit-requests/analytics`	Get mentor review analytics	Yes	Mentor
GET	`/api/mentors/skill-gaps`	Get skill gap analysis	Yes	Mentor
GET	`/api/mentors/department/performance`	Get department performance metrics	Yes	Mentor
POST	`/api/mentors/interventions`	Create student intervention	Yes	Mentor

GET	`/api/mentors/interventions`	Get all interventions	Yes	Mentor
GET	`/api/mentor/analytics`	Get mentor-specific analytics	Yes	Mentor
GET	`/api/mentor/analytics/department`	Get department analytics	Yes	Mentor

Mentor Workflow Mapping:

This section maps each step in the mentor journey (see Mentor Journey diagram) to the specific API endpoints used.

1. Dashboard & Overview

Workflow Steps: Mentor Login → View Dashboard → Select Task Type

Step	Endpoint	Method	Description
Login	`/api/auth/login`	POST	Authenticate with Firebase token
View dashboard	`/api/mentors/dashboard`	GET	View dashboard with pending tasks and metrics
View profile	`/api/mentor/profile`	GET	View mentor profile and workload
Update profile	`/api/mentor/profile`	PUT	Update mentor information and preferences
Update account info	`/api/auth/me`	PATCH	Update email and account settings

2. Student Management

Workflow Steps: View and manage assigned students

Step	Endpoint	Method	Description
View assigned students	`/api/mentor/students/list`	GET	List all assigned students with filters
View student details	`/api/mentor/students/:studentId/details`	GET	View student profile, history, and performance
View student applications	`/api/mentor/students/:studentId/applications`	GET	View all applications from specific student
Track student progress	`/api/mentors/students/:studentId/progress`	GET	View detailed progress tracking for student
Create intervention	`/api/mentors/interventions`	POST	Create intervention for

			struggling student
View interventions	`/api/mentors/interventions`	GET	View all student interventions
View assigned students (legacy)	`/api/mentors/students`	GET	Legacy endpoint for assigned students

3. Application Review

Workflow Steps: Review Applications Queue → View Application Details → Check Student Profile → Review AI Match Score → Approve/Reject

Step	Endpoint	Method	Description
View pending applications	`/api/mentors/applications/pending`	GET	Get queue of applications awaiting mentor review
View application details	`/api/mentors/applications/:applicationId`	GET	View full application with student profile and AI ranking
Approve application	`/api/mentors/applications/:applicationId/approve`	POST	Approve application and forward to company
Reject application	`/api/mentors/applications/:applicationId/reject`	POST	Reject application with constructive feedback

4. Internship Approval

Workflow Steps: Review Internship Postings → View Internship Details → Verify Department Fit → Approve/Reject

Step	Endpoint	Method	Description
View pending internships	`/api/mentor/internships/pending`	GET	Get queue of internships awaiting mentor approval
View all internships	`/api/mentor/internships`	GET	List all internships with filtering options
View internship details	`/api/mentor/internships/:internshipId`	GET	View full internship details for review
Approve internship	`/api/mentor/internships/:internshipId/approve`	POST	Approve internship for department students
Reject internship	`/api/mentor/internships/:internshipId/reject`	POST	Reject internship with

			reasons and feedback

5. Logbook Review

Workflow Steps: Review logbook entries from students during internships

Step	Endpoint	Method	Description
View pending logbooks	`/api/mentors/logbooks/pending`	GET	Get queue of logbooks awaiting review
View logbook details	`/api/mentors/logbooks/:logbookId`	GET	View full logbook entry with attachments
Approve logbook	`/api/mentors/logbooks/:logbookId/approve`	POST	Approve logbook entry
Request revision	`/api/mentors/logbooks/:logbookId/revision`	POST	Request revisions to logbook entry

6. Credit Request Review

Workflow Steps: Review Credit Requests → Review Logbook → Check Quality Criteria → Approve/Request Revisions

Step	Endpoint	Method	Description
View pending credit requests	`/api/mentors/:mentorId/credit-requests/pending`	GET	Get queue of credit requests awaiting mentor review
View credit request details	`/api/mentors/:mentorId/credit-requests/:requestId`	GET	View full credit request with completion details
Submit review decision	`/api/mentors/:mentorId/credit-requests/:requestId/review`	POST	Approve or reject credit request with comments
View review history	`/api/mentors/:mentorId/credit-requests/history`	GET	View history of all credit request reviews
View review analytics	`/api/mentors/:mentorId/credit-requests/analytics`	GET	View analytics on credit request reviews
View pending credits (legacy)	`/api/mentors/credits/pending`	GET	Legacy endpoint for pending credit requests
Decide on credit (legacy)	`/api/mentors/credits/:requestId/decide`	POST	Legacy endpoint for credit decision

7. Analytics & Reporting

Workflow Steps: View performance metrics and insights

Step	Endpoint	Method	Description
View mentor analytics	`/api/mentor/analytics`	GET	View personal mentoring analytics and metrics
View department	`/api/mentor/analytics/department`	GET	View department-wide

analytics			analytics
View department performance	`/api/mentors/department/performance`	GET	View detailed department performance metrics
View skill gap analysis	`/api/mentors/skill-gaps`	GET	View skill gap analysis for students

8. Notifications

Workflow Steps: Receive and manage notifications throughout all workflows

Step	Endpoint	Method	Description
View notifications	`/api/notifications`	GET	Get paginated list of notifications
Mark notification as read	`/api/notifications/:id/read`	PATCH	Mark specific notification as read
Mark all as read	`/api/notifications/read-all`	PATCH	Mark all notifications as read

Admin Endpoints

HTTP Method	Endpoint	Description	Auth Required	Permissions
GET	`/api/admin/dashboard`	Get admin dashboard with metrics	Yes	Admin
GET	`/api/admin/companies`	Get companies pending verification	Yes	Admin
GET	`/api/companies/:id`	Get company details	Yes	Admin
PUT	`/api/admin/companies/:id/verify`	Verify/reject company	Yes	Admin
PUT	`/api/admin/companies/:id/status`	Update company status	Yes	Admin
GET	`/api/admin/internships`	Get internships pending approval	Yes	Admin
GET	`/api/internships/:id`	Get internship details	Yes	Admin
PUT	`/api/admin/internships/:id/review`	Approve/reject internship	Yes	Admin
GET	`/api/admin/applications`	Get all applications	Yes	Admin
GET	`/api/admin/credit-requests`	Get credit requests for review	Yes	Admin
GET	`/api/credit-requests/:id`	Get credit request details	Yes	Admin
PUT	`/api/admin/credit-requests/:id/review`	Final approval/rejection	Yes	Admin
GET	`/api/admin/students`	Get all students	Yes	Admin

GET	`/api/admin/mentors`	Get all mentors	Yes	Admin
POST	`/api/admin/mentors`	Create mentor account	Yes	Admin (super)
PUT	`/api/admin/mentors/:id`	Update mentor details	Yes	Admin (super)
GET	`/api/admin/analytics`	Get system analytics	Yes	Admin
POST	`/api/admin/analytics/export`	Export analytics reports	Yes	Admin
GET	`/api/admin/audit-logs`	Get audit logs	Yes	Admin (super)
GET	`/api/notifications`	Get notifications	Yes	Admin

Admin Workflow Mapping:

This section maps each step in the admin journey (see Admin Journey diagram) to the specific API endpoints used.

1. Dashboard & Overview

Workflow Steps: Admin Login → View Dashboard → Select Task Type

Step	Endpoint	Method	Description
Login	`/api/auth/login`	POST	Authenticate with Firebase token
View dashboard	`/api/admin/dashboard`	GET	View system-wide dashboard with pending tasks and metrics
Update account info	`/api/auth/me`	PATCH	Update email and account settings

2. Company Verification

Workflow Steps: Review Company Verifications → View Company Details → Review AI Verification → Check Documents → Verify/Reject

Step	Endpoint	Method	Description
View companies pending verification	`/api/admin/companies`	GET	List companies awaiting verification with filters
View company details	`/api/companies/:id`	GET	View full company details, documents, and AI verification
Verify or reject company	`/api/admin/companies/:id/verify`	PUT	Approve or reject company with detailed feedback
Update company status	`/api/admin/companies/:id/status`	PUT	Update company status (suspend, block, reactivate)

3. Internship Approval

Workflow Steps: Review Internship Postings → View Internship Details → Verify Compliance → Approve/Reject

Step	Endpoint	Method	Description
View internships pending approval	`/api/admin/internships`	GET	List internships awaiting admin approval
View internship details	`/api/internships/:id`	GET	View full internship details and company verification
Approve or reject internship	`/api/admin/internships/:id/review`	PUT	Approve for mentor review or reject with feedback

4. Credit Request Final Approval

Workflow Steps: Review Credit Requests → Review Mentor Approval → Check Compliance → Final Approval/Rejection

Step	Endpoint	Method	Description
View credit requests for review	`/api/admin/credit-requests`	GET	List credit requests awaiting final admin approval
View credit request details	`/api/credit-requests/:id`	GET	View full credit request with mentor review
Final approval or rejection	`/api/admin/credit-requests/:id/review`	PUT	Perform final approval and trigger credit addition

5. User Management

Workflow Steps: Manage system users and permissions

Step	Endpoint	Method	Description
View all students	`/api/admin/students`	GET	List all students with filters and search
View all mentors	`/api/admin/mentors`	GET	List all mentors with workload info
Create mentor account	`/api/admin/mentors`	POST	Create new mentor account (super admin only)
Update mentor details	`/api/admin/mentors/:id`	PUT	Update mentor information (super admin only)
View all applications	`/api/admin/applications`	GET	View all applications across the system

6. Analytics & Reporting

Workflow Steps: View System Analytics → Export Reports

Step	Endpoint	Method	Description
View system analytics	`/api/admin/analytics`	GET	View comprehensive system analytics with date ranges
Export analytics reports	`/api/admin/analytics/export`	POST	Export analytics as CSV or PDF reports
View audit logs	`/api/admin/audit-logs`	GET	View system audit logs (super admin

			only)

7. Notifications

Workflow Steps: Receive and manage notifications throughout all workflows

Step	Endpoint	Method	Description
View notifications	`/api/notifications`	GET	Get paginated list of notifications
Mark notification as read	`/api/notifications/:id/read`	PATCH	Mark specific notification as read
Mark all as read	`/api/notifications/read-all`	PATCH	Mark all notifications as read

Common Request/Response Formats

All API responses follow consistent formats for success, error, and paginated data.

Success Response:

```
{
  "success": true,
  "data": {
    // Response data (object or array)
  },
  "message": "Operation successful"
}
```

Example - Get Student Profile:

```
{
  "success": true,
  "data": {
    "studentId": "STU-2024-001",
    "email": "student@example.com",
    "profile": {
      "name": "John Doe",
      "department": "Computer Science"
    },
    "readinessScore": 85,
    "credits": {
      "earned": 12,
      "pending": 4
    }
  },
  "message": "Profile retrieved successfully"
}
```

Error Response:

```
{
  "success": false,
  "error": {
    "code": "ERROR_CODE",
```

```
    "message": "Human-readable error message",
    "details": {}
  }
}
```

Example - Validation Error:

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid request data",
    "details": {
      "fields": {
        "email": "Email is required",
        "coverLetter": "Cover letter must be at least 100 characters"
      }
    }
  }
}
```

Pagination Response:

```
{
  "success": true,
  "data": {
    "items": [],
    "pagination": {
      "page": 1,
      "limit": 20,
      "total": 100,
      "pages": 5
    }
  }
}
```

Example - Paginated Internships:

```
{
  "success": true,
  "data": {
    "items": [
      {
        "internshipId": "INT-2024-001",
        "title": "Software Engineering Intern",
        "companyName": "Tech Corp",
        "department": "Computer Science",
        "slotsRemaining": 3
      }
    ],
    "pagination": {
      "page": 1,
      "limit": 20,
```

```
      "total": 45,
      "pages": 3
    }
  },
  "message": "Internships retrieved successfully"
}
```

Common Error Codes

Error Code	HTTP Status	Description	Resolution
`AUTH_REQUIRED`	401	Authentication token missing or invalid	Provide valid Firebase authentication token in Authorization header
`FORBIDDEN`	403	User lacks permission for this action	Check user role and permissions for the requested endpoint
`NOT_FOUND`	404	Requested resource doesn't exist	Verify the resource ID exists and user has access
`VALIDATION_ERROR`	400	Request data validation failed	Check request body against API documentation and fix validation errors
`STATE_TRANSITION_ERROR`	400	Invalid status transition attempted	Review state transition diagrams for valid transitions
`DUPLICATE_ENTRY`	409	Resource already exists	Check if resource already exists before creating
`QUOTA_EXCEEDED`	429	Rate limit or quota exceeded	Implement exponential backoff and retry logic
`INTERNAL_ERROR`	500	Internal server error	Contact support if error persists

API Best Practices

Authentication:

- Always include Firebase authentication token in `Authorization` header
- Format: `Bearer <firebase_token>`
- Tokens expire after 1 hour - implement token refresh logic

Error Handling:

- Always check the `success` field in responses
- Parse error codes and display user-friendly messages
- Log error details for debugging

Pagination:

- Use `page` and `limit` query parameters for list endpoints
- Default limit is 20, maximum is 100
- Always check `pagination.total` to determine if more pages exist

Rate Limiting:

- Implement exponential backoff for rate limit errors

- Cache frequently accessed data on the client side
- Batch operations when possible

Status Polling:

- Use WebSocket connections for real-time updates when available
- For polling, use reasonable intervals (30-60 seconds minimum)
- Implement exponential backoff for polling

Workflow-to-Endpoint Cross-Reference

This section provides a comprehensive mapping of all major workflows to their corresponding API endpoints, organized by workflow type.

Application Approval Workflow

Flow: Student → Mentor → Company → Student → Completion

Workflow Step	Role	Endpoint	Method	Descripti
1. Submit application	Student	`/api/students/internships/:internshipId/apply`	POST	Student applies to internship
2. View pending applications	Mentor	`/api/mentors/applications/pending`	GET	Mentor sees pending applicatio
3. Review application	Mentor	`/api/mentors/applications/:applicationId`	GET	Mentor views applicatio details
4. Approve application	Mentor	`/api/mentors/applications/:applicationId/approve`	POST	Mentor approves and forwards company
5. Receive notification	Student	`/api/notifications`	GET	Student notified o mentor approval
6. View applicants	Company	`/api/companies/internships/:internshipId/applicants`	GET	Company views approved applicant
7. Review application	Company	`/api/companies/applications/:applicationId`	GET	Company reviews applicatio details

8. Shortlist candidate	Company	`/api/companies/applications/approve`	POST	Company shortlists candidate
9. Receive notification	Student	`/api/notifications`	GET	Student notified c shortlistin
10. Check application status	Student	`/api/students/applications/:applicationId`	GET	Student views updated status
11. Accept offer	Student	(Implicit in status)	-	Student accepts through applicatic status
12. Mark completion	Company	`/api/companies/completions/:completionId/mark-complete`	PUT	Company marks internship complete

Credit Transfer Workflow

Flow: Completion → Mentor → Admin → Credits Added

Workflow Step	Role	Endpoint	Method	Des
1. Mark completion	Company	`/api/companies/completions/:completionId/mark-complete`	PUT	Comp intern comp
2. Auto-create credit request	System	(Automatic)	-	Syster credit
3. View completed internships	Student	`/api/students/internships/completed`	GET	Stude comp
4. Create credit request	Student	`/api/students/:studentId/credit-requests`	POST	Stude credit
5. View pending requests	Mentor	`/api/mentors/:mentorId/credit-requests/pending`	GET	Ment pendi reque
6. Review credit request	Mentor	`/api/mentors/:mentorId/credit-requests/:requestId`	GET	Ment detail

7. Submit mentor review	Mentor	`/api/mentors/:mentorId/credit-requests/:requestId/review`	POST	Mente appro
8. Receive notification	Student	`/api/notifications`	GET	Stude of me decisi
9. Resubmit if rejected	Student	`/api/students/:studentId/credit-requests/:requestId/resubmit`	PUT	Stude resub chang
10. View pending admin reviews	Admin	`/api/admin/credit-requests`	GET	Admi mente appro reque
11. Review credit request	Admin	`/api/credit-requests/:id`	GET	Admi detail
12. Final approval	Admin	`/api/admin/credit-requests/:id/review`	PUT	Admi final a
13. Credits added	System	(Automatic)	-	Syster credit stude
14. View credit history	Student	`/api/students/:studentId/credits/history`	GET	Stude updat
15. Download certificate	Student	`/api/students/:studentId/credits/certificate/:requestId`	GET	Stude down certifi

Internship Posting Workflow

Flow: Company → Admin → Mentor → Live

Workflow Step	Role	Endpoint	Method	Description
1. Create internship	Company	`/api/companies/internships`	POST	Company creates posting
2. View pending internships	Admin	`/api/admin/internships`	GET	Admin sees pending postings
3. Review internship	Admin	`/api/internships/:id`	GET	Admin reviews details
4. Approve internship	Admin	`/api/admin/internships/:id/review`	PUT	Admin approves

				for mentor review
5. Receive notification	Company	`/api/notifications`	GET	Company notified of admin approval
6. View pending internships	Mentor	`/api/mentor/internships/pending`	GET	Mentor sees pending approvals
7. Review internship	Mentor	`/api/mentor/internships/:internshipId`	GET	Mentor reviews details
8. Approve internship	Mentor	`/api/mentor/internships/:internshipId/approve`	POST	Mentor approves for students
9. Receive notification	Company	`/api/notifications`	GET	Company notified internship is live
10. Browse internships	Student	`/api/students/internships`	GET	Students can now see posting

Company Verification Workflow

Flow: Registration → AI Check → Admin Review → Verified/Rejected

Workflow Step	Role	Endpoint	Method	Description
1. Register company	Company	`/api/auth/companies/register`	POST	Company creates account
2. Upload documents	Company	`/api/companies/documents`	POST	Company uploads verification docs
3. AI verification	System	(Automatic)	-	AI analyzes documents
4. Check status	Company	`/api/companies/status`	GET	Company checks verification status
5. View pending companies	Admin	`/api/admin/companies`	GET	Admin sees pending verifications
6. Review company	Admin	`/api/companies/:id`	GET	Admin reviews

				details and AI results
7. Verify or reject	Admin	`/api/admin/companies/:id/verify`	PUT	Admin makes decision
8. Receive notification	Company	`/api/notifications`	GET	Company notified of decision
9. Submit reappeal (if rejected)	Company	`/api/companies/reappeal`	POST	Company submits reappeal
10. Check reappeal status	Company	`/api/companies/reappeal/status`	GET	Company checks reappeal status
11. Review reappeal	Admin	`/api/companies/:id`	GET	Admin reviews reappeal
12. Decide on reappeal	Admin	`/api/admin/companies/:id/verify`	PUT	Admin approves or blocks

Student Learning & Development Workflow

Flow: Browse Modules → Complete Training → Practice Interviews → Improve Readiness

Workflow Step	Role	Endpoint	Method	Description
1. View dashboard	Student	`/api/students/dashboard`	GET	Student sees readiness score
2. Get recommendations	Student	`/api/students/modules/recommended`	GET	Get personalized module recommendations
3. Start module	Student	`/api/students/modules/start`	POST	Begin learning module
4. Complete module	Student	`/api/students/modules/complete`	POST	Complete module and update score
5. Start interview practice	Student	`/api/students/interviews/start`	POST	Begin AI interview practice
6. Submit answers	Student	`/api/students/interviews/answer`	POST	Submit answers for AI feedback
7. End interview	Student	`/api/students/interviews/end`	POST	Complete interview session
8. View history	Student	`/api/students/interviews/history`	GET	Review past practice sessions
9. Get AI guidance	Student	`/api/students/chatbot`	POST	Ask AI for career guidance

10. View updated profile	Student	`/api/students/profile`	GET	See updated readiness score

Logbook Management Workflow

Flow: Student Submits → Company Reviews → Mentor Reviews

Workflow Step	Role	Endpoint	Method	Description
1. Submit logbook entry	Student	`/api/students/logbooks`	POST	Student submits daily/weekly entry
2. View intern logbooks	Company	`/api/companies/interns/:studentId/logbooks`	GET	Company views logbook entries
3. Provide feedback	Company	`/api/companies/logbooks/:logbookId/feedback`	POST	Company provides feedback
4. View pending logbooks	Mentor	`/api/mentors/logbooks/pending`	GET	Mentor sees pending reviews
5. Review logbook	Mentor	`/api/mentors/logbooks/:logbookId`	GET	Mentor reviews entry
6. Approve or request revision	Mentor	`/api/mentors/logbooks/:logbookId/approve`	POST	Mentor approves or requests changes
7. Receive notification	Student	`/api/notifications`	GET	Student notified of feedback
8. View logbooks	Student	`/api/students/logbooks`	GET	Student sees feedback and status

---

Quick Reference

Entity Quick Reference

Student

- **ID Format**: `STU-YYYY-NNN`
- **Key Fields**: studentId, email, profile, readinessScore, credits
- **Status Values**: active, inactive, suspended
- **Relationships**: Applications, CreditRequests, InternshipCompletions, Mentor

Company

- **ID Format**: `COM-YYYY-NNN`
- **Key Fields**: companyId, companyName, email, status, documents
- **Status Values**: pending_verification, verified, rejected, blocked, reappeal, suspended
- **Relationships**: Internships, Applications

Mentor

- **ID Format**: `MEN-YYYY-NNN`
- **Key Fields**: mentorId, email, profile, assignedStudents, workload
- **Status Values**: active, inactive, on_leave
- **Relationships**: Students, Applications, CreditRequests

Admin

- **ID Format**: `ADM-YYYY-NNN`
- **Key Fields**: adminId, email, name, role, permissions
- **Roles**: super_admin, admin, support
- **Status Values**: active, inactive

Internship

- **ID Format**: `INT-YYYY-NNN`
- **Key Fields**: internshipId, companyId, title, department, status, slots
- **Status Values**: draft, pending_admin_verification, admin_approved, pending_mentor_verification, open_for_applications, closed, cancelled
- **Relationships**: Company, Applications, InternshipCompletions

Application

- **ID Format**: `APP-YYYY-NNN`
- **Key Fields**: applicationId, studentId, internshipId, status, aiRanking
- **Status Values**: pending, mentor_approved, mentor_rejected, shortlisted, rejected, accepted, withdrawn, completed
- **Relationships**: Student, Internship, Company, InternshipCompletion

InternshipCompletion

- **ID Format**: `CMP-YYYY-NNN`
- **Key Fields**: completionId, studentId, internshipId, totalHours, creditsEarned
- **Status Values**: pending, completed, verified
- **Relationships**: Student, Internship, Application, CreditRequest

CreditRequest

- **ID Format**: `CR-YYYY-NNN`
- **Key Fields**: creditRequestId, studentId, internshipCompletionId, requestedCredits, status
- **Status Values**: pending_mentor_review, mentor_approved, mentor_rejected, pending_admin_review, admin_approved, admin_rejected, pending_student_action, credits_added, completed
- **Relationships**: Student, InternshipCompletion, Mentor

Notification

- **ID Format**: `NOT-YYYY-NNN`
- **Key Fields**: notificationId, userId, role, type, title, message, priority, read
- **Priority Values**: low, medium, high, critical
- **Role Values**: student, mentor, admin, company
- **Delivery Channels**: email, sms, whatsapp, push
- **Relationships**: Student, Company, Mentor, Admin (polymorphic via userId and role)

Status Values

Internship Status

1. `draft` - Initial creation by company
2. `pending_admin_verification` - Submitted for admin review
3. `admin_approved` - Admin approved, awaiting mentor
4. `admin_rejected` - Rejected by admin
5. `pending_mentor_verification` - Awaiting mentor approval
6. `mentor_rejected` - Rejected by mentor
7. `open_for_applications` - Live and accepting applications
8. `closed` - Deadline passed or slots filled
9. `cancelled` - Cancelled by company

Application Status

1. `pending` - Submitted, awaiting mentor review
2. `mentor_approved` - Mentor approved, sent to company
3. `mentor_rejected` - Rejected by mentor
4. `shortlisted` - Company shortlisted candidate
5. `rejected` - Rejected by company
6. `accepted` - Student accepted offer
7. `withdrawn` - Student withdrew application
8. `completed` - Internship completed

Credit Request Status

1. `pending_mentor_review` - Awaiting mentor review
2. `mentor_approved` - Mentor approved, sent to admin
3. `mentor_rejected` - Rejected by mentor
4. `pending_student_action` - Needs student revision
5. `pending_admin_review` - Awaiting admin review
6. `admin_approved` - Admin approved
7. `admin_rejected` - Rejected by admin
8. `credits_added` - Credits added to account
9. `completed` - Process complete with certificate

Company Verification Status

1. `pending_verification` - Awaiting verification
2. `verified` - Verified and active
3. `rejected` - Verification rejected
4. `reappeal` - Reappeal submitted
5. `blocked` - Permanently blocked
6. `suspended` - Temporarily suspended

Role Permissions Matrix

Action	Student	Company	Mentor	Admin	Super Admin
Authentication					
Register	✓	✓	-	-	-

Login	✓	✓	✓	✓	✓
Update Own Profile	✓	✓	✓	✓	✓
Internships					
Browse Internships	✓	-	✓	✓	✓
View Internship Details	✓	✓	✓	✓	✓
Create Internship	-	✓	-	-	-
Update Internship	-	✓	-	-	-
Delete Internship	-	✓	-	-	-
Approve Internship (Admin)	-	-	-	✓	✓
Approve Internship (Mentor)	-	-	✓	-	-
Applications					
Submit Application	✓	-	-	-	-
View Own Applications	✓	-	-	-	-
Withdraw Application	✓	-	-	-	-
Accept Offer	✓	-	-	-	-
Review Application (Mentor)	-	-	✓	-	-
Review Application (Company)	-	✓	-	-	-
View All Applications	-	-	-	✓	✓
Credit Requests					
Submit Credit Request	✓	-	-	-	-
View Own Credit Requests	✓	-	-	-	-
Update Credit Request	✓	-	-	-	-
Review Credit (Mentor)	-	-	✓	-	-
Review Credit (Admin)	-	-	-	✓	✓
View All Credit Requests	-	-	-	✓	✓
Company Management					
Upload Documents	-	✓	-	-	-
Submit Reappeal	-	✓	-	-	-
Verify Company	-	-	-	✓	✓
Update Company Status	-	-	-	✓	✓
View All Companies	-	-	-	✓	✓
Completion Management					

View Own Completions	✓	-	-	-	-
Mark Completion	-	✓	-	-	-
Submit Evaluation	-	✓	-	-	-
User Management					
View Students	-	-	✓	✓	✓
View Mentors	-	-	-	✓	✓
Create Mentor	-	-	-	-	✓
Update Mentor	-	-	-	-	✓
Analytics					
View Own Analytics	✓	✓	✓	-	-
View System Analytics	-	-	-	✓	✓
Export Reports	-	-	-	✓	✓
View Audit Logs	-	-	-	-	✓
Notifications					
View Own Notifications	✓	✓	✓	✓	✓
Mark Notification Read	✓	✓	✓	✓	✓

Legend:

- ✓ = Permission granted
- - = Permission not granted

Common Workflow Patterns

Application Approval Workflow

```
Student → Mentor → Company → Student → Completion
```

1. Student submits application
2. Mentor reviews and approves/rejects
3. Company reviews approved applications
4. Student accepts offer
5. Company marks completion

Credit Transfer Workflow

```
Completion → Mentor → Admin → Credits Added
```

1. System creates credit request on completion
2. Mentor reviews logbook and quality
3. Admin performs final compliance check
4. System adds credits and generates certificate

Internship Posting Workflow

```
Company → Admin → Mentor → Live
```

1. Company creates internship posting
2. Admin verifies compliance
3. Mentor verifies department fit
4. Internship becomes available to students

Company Verification Workflow

```
Registration → AI Check → Admin Review → Verified/Rejected
```

1. Company registers and uploads documents
2. AI performs initial verification
3. Admin reviews AI results and documents
4. Company verified or rejected (with reappeal option)

Appendix

External Service Integrations

Firebase Authentication:

- User authentication and session management
- Role-based access control
- Token verification

Cloudflare R2 Storage:

- Document storage (company documents, certificates)
- Profile image storage
- Secure file access with signed URLs

Email Service:

- Notification emails
- Status update emails
- Reminder emails

Gemini AI:

- Company verification analysis
- Application ranking and matching
- Skill gap analysis
- Document analysis

Database Indexes

Key indexes for performance:

- Student: `studentId`, `firebaseUid`, `email`
- Company: `companyId`, `firebaseUid`, `email`, `status`
- Internship: `internshipId`, `companyId`, `status`, `department`
- Application: `applicationId`, `studentId`, `internshipId`, `status`
- CreditRequest: `creditRequestId`, `studentId`, `status`

Caching Strategy

Redis caching is used for:

- User session data
- Frequently accessed internship listings
- Dashboard metrics
- Application counts
- Notification counts

Cache TTL varies by data type (5 minutes to 1 hour).

Requirements Coverage

This section verifies that all requirements from the requirements document are covered in this documentation.

Requirement 1: Entity Relationship Diagrams ✓

- **1.1** Core entities displayed: Student, Company, Mentor, Admin, Internship, Application, CreditRequest, InternshipCompletion, Notification ✓
- **1.2** Relationships shown with cardinality indicators (1:1, 1:N, N:M) ✓
- **1.3** Key fields displayed including primary keys, foreign keys, and business fields ✓
- **1.4** Standard ERD notation with clear visual distinction ✓

Location: Entity Relationship Diagram

Requirement 2: User Flow Diagrams ✓

- **2.1** Student flow from registration through credit request ✓
- **2.2** Company flow from registration through intern management ✓
- **2.3** Mentor flow for application approval and credit review ✓
- **2.4** Admin flow for verification and approvals ✓
- **2.5** Consistent visual notation with decision points ✓

Location: User Workflows

Requirement 3: API Endpoint Mappings ✓

- **3.1** REST API endpoints organized by user role ✓
- **3.2** HTTP method, URL path, authentication requirements, and descriptions ✓
- **3.3** Workflow steps mapped to corresponding API endpoints ✓
- **3.4** Request/response examples for complex endpoints ✓
- **3.5** Role-based permissions indicated for each endpoint ✓

Location: API Endpoint Reference, Workflow-to-Endpoint Cross-Reference

Requirement 4: System Architecture ✓

- **4.1** Frontend, backend, database, and external service layers displayed ✓
- **4.2** Data flow between layers with directional arrows ✓
- **4.3** External integrations (Firebase, storage, notifications) displayed ✓
- **4.4** Clear visual hierarchy distinguishing architectural layers ✓

Location: System Architecture

Requirement 5: State Transition Diagrams ✓

- **5.1** Internship status transitions from draft through completion ✓
- **5.2** Application status transitions through approval workflow ✓
- **5.3** Credit request status transitions through multi-stage approval ✓
- **5.4** Company verification status transitions including reappeal ✓
- **5.5** Roles indicated for each state transition ✓

Location: State Transition Diagrams

Document Version

Version: 1.0
Last Updated: December 5, 2024
Maintained By: Development Team

Change Log:

- v1.0 (December 2024): Initial comprehensive documentation release
  - Complete entity relationship diagrams
  - All user workflow diagrams
  - Comprehensive API endpoint reference
  - State transition diagrams
  - Quick reference guides

For questions or updates to this documentation, please contact the development team.

Feedback and Contributions

This documentation is a living document. If you find:

- Errors or inconsistencies
- Missing information
- Outdated content
- Areas needing clarification

Please submit feedback to the development team or create an issue in the project repository.